THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A mixture of oil and water is separated using: 1

(a) sieve **(b)** magnet
(c) a separating funnel **(d)** filter paper

2. 100 g of a 10 per cent solution is mixed with 100 g of water. The new mass percentage is: 1

(a) 5 per cent **(b)** 10 per cent **(c)** 20 per cent **(d)** 2.5 per cent

3. Air bubbles are seen when water is heated before boiling. This is due to: 1

(a) dissolved air coming out **(b)** steam
(c) hydrogen **(d)** salt

4. Ink on paper is dissolved by: 1

(a) stone **(b)** alcohol or water **(c)** air **(d)** sand

5. 50 g of water is evaporated from 200 g of a 10 per cent solution. The new mass percentage is: 1

(a) 5 per cent **(b)** 10 per cent **(c)** 20 per cent **(d)** 13.3 per cent

6. Which of the following methods can separate sugar from sugar solution? 1

(a) magnet **(b)** evaporation **(c)** filtration **(d)** sieve

7. Rate of dissolving of a solid depends on: 1

(a) its colour **(b)** smell
(c) the size of the beaker **(d)** surface area of the solid

8. To obtain a saturated solution of sugar at a certain temperature, 200 g of sugar is dissolved in 100 g of water. The solubility is: 1

(a) 100 g per 100 g of water **(b)** 200 g per 100 g of water
(c) 300 g per 100 g of water **(d)** 66 g per 100 g of water

9. Which solvent dissolves nail polish? 1

(a) water **(b)** milk **(c)** salt solution **(d)** acetone

10. A powder of copper sulphate dissolves in hot water fully but not cold water. This shows: 1

(a) solubility increases with temperature **(b)** powder is magnetic
(c) water is heavy **(d)** none

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does cold water hold more dissolved oxygen than warm water? 2

12. Why is it easier to dissolve sugar in hot water than in cold water? 2

13. A solution contains 4 g of solute in 96 g of solvent. Find the mass percentage. 2

14. What is a dilute acid? How is it made from a concentrated one? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why a solution is transparent but a suspension is not. 3

16. Explain the saturated solution as a dynamic state at the particle level. 3

17. Why is sea water saltier in hot, dry regions? Explain with a concept from this chapter. 3

OR

Compare solution, suspension and colloid with respect to particle size and Tyndall effect. 3

18. Explain the effect of pressure on the solubility of gases using soda water. 3

OR

Why do nail polish and paint need special solvents such as acetone or turpentine? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe how solutions are used in the kitchen, medicine, industry and agriculture with two examples for each. 5

OR

Discuss the solubility of common substances: sugar, salt, chalk, oil, alcohol, giving reasons. 5

20. Design a fair test to compare the rate of dissolving of sugar crystals and powdered sugar in water at the same temperature. 5

OR

Write a short note on concentration of solutions, explaining concentrated, dilute and mass percentage with an example calculation. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND